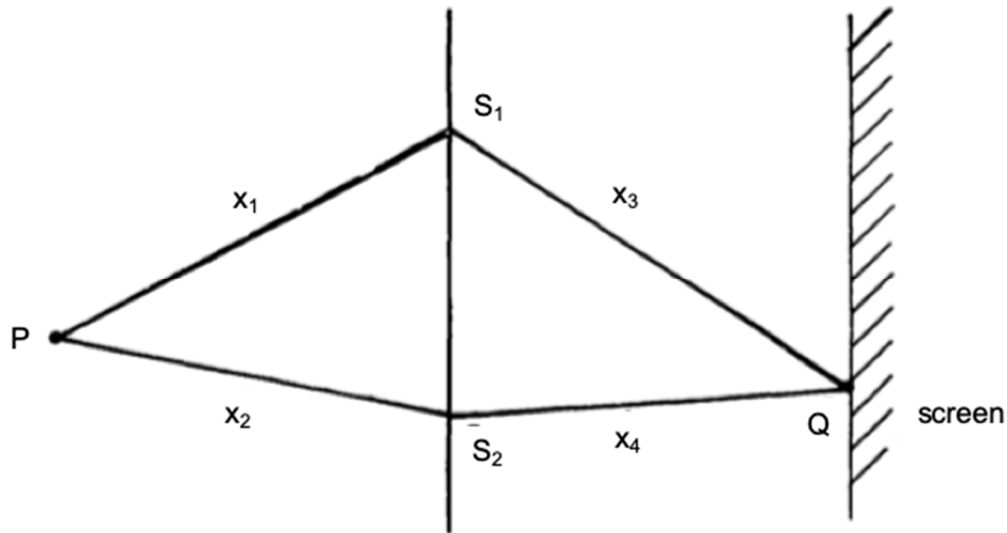


magnitude, which is x_0 . Therefore, Option D is wrong.

- 19 Two identical narrow slits S_1 and S_2 are illuminated with light of wavelength λ from a point source P.



If, as shown in the diagram above, the light is then allowed to fall on a screen, and if m is a positive integer, the condition for constructive interference at Q is

- | | |
|---|--|
| A | $x_1 - x_2 = (2m+1) \frac{\lambda}{2}$ |
| B | $x_3 - x_4 = (2m+1) \frac{\lambda}{2}$ |
| C | $x_3 - x_4 = m\lambda$ |
| D | $(x_1 + x_3) - (x_2 + x_4) = m\lambda$ |

Answer: D

	For constructive interference to take place, the waves meeting at the point must have a path difference (difference in the lengths of paths from the source to the meeting point) to be a full wavelength, or integers of multiples of the wavelengths of the waves.	
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